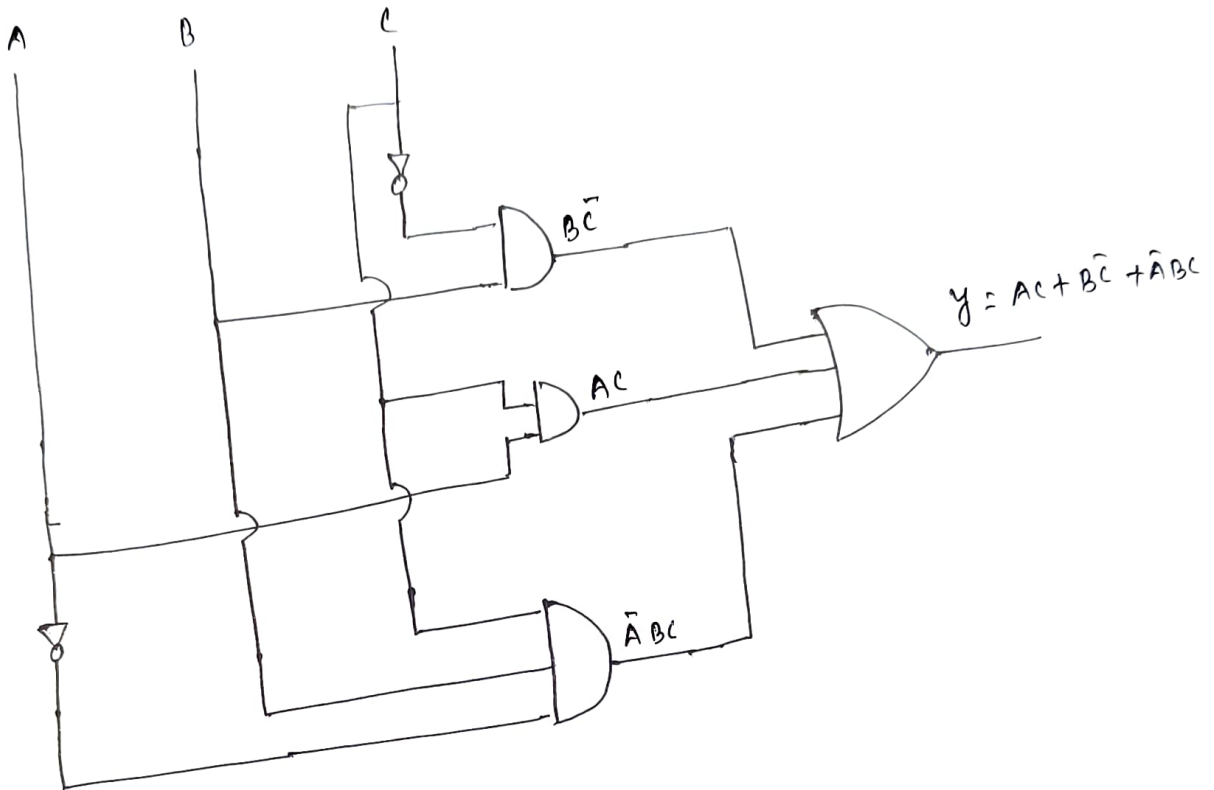


1. $y = AC + B\bar{C} + \bar{A}BC$



2. Given $F = AB + A(B+C) + B(B+C)$

$$\begin{aligned}
 &= AB + AB + AC + B + BC \\
 &= AB + B + AC + BC \\
 &= B(A+1) + AC + BC \\
 &= B + BC + AC \\
 &= B(1+C) + AC \\
 &= \boxed{B + AC}
 \end{aligned}$$

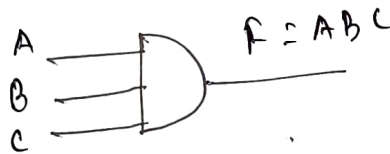
③ Given $F = ((AB)' + (AC)')'$

using Demorgan's law

$$F = ((AB)')' \cdot ((AC)')'$$

$$F = AB \cdot AC$$

$$F = ABC$$



4. ① Given $A \cdot (\overline{A+B})$

$$= A \cdot (\overline{A} \cdot \overline{B}) = 0$$

② $AB + A\overline{B}$

$$A(B + \overline{B}) = A$$

③ $AC + AB\overline{C} + A\overline{B}C$

$$= AC(1 + \overline{B}) + A\overline{B}C$$

$$= \boxed{AC + A\overline{B}C} \Rightarrow \text{It is one of the solution.}$$

It means the term $A\overline{B}C$ is redundant given in the question.

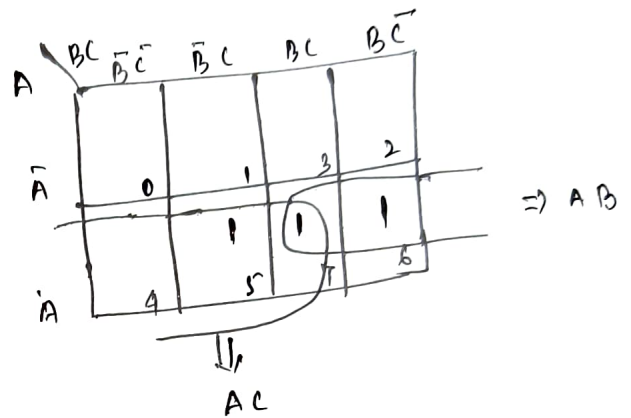
Now, considering the solⁿ $AC + AB\bar{C}$

$$AC = A(B + \bar{B}) \cdot C = ABC + AB\bar{C}$$

$$\text{So, } AC + AB\bar{C} = \underset{(7)}{ABC} + \underset{(5)}{AB\bar{C}} + \underset{(6)}{AB\bar{C}}$$

Forming 3 variable input K-map

So, Grouping all 1's in efficient manner, we get $(AB + AC)$ as output



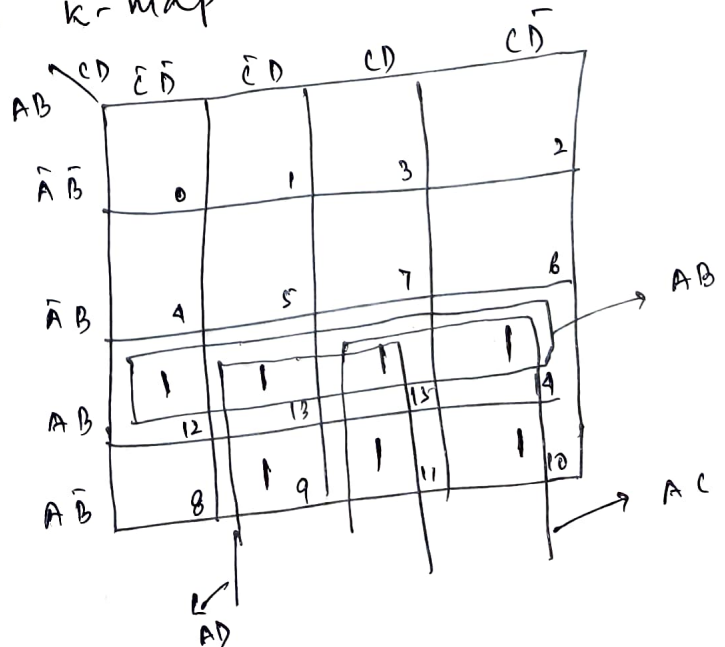
So, $(AC + AB\bar{C} + AB\bar{C})$ can be reduced in $(AB + AC)$

d) Given

$$\underset{(12)}{ABC'D'} + \underset{(13)}{ABC'D} + \underset{(9)}{AB'C'D} + \underset{(15)}{ABCD} + \underset{(11)}{AB'C'D} + \underset{(14)}{ABCD'} + \underset{(10)}{AB'C'D'}$$

Using 4 variable K-map

$$\text{So, O/P} = AB + AC + AD = A(B + C + D)$$

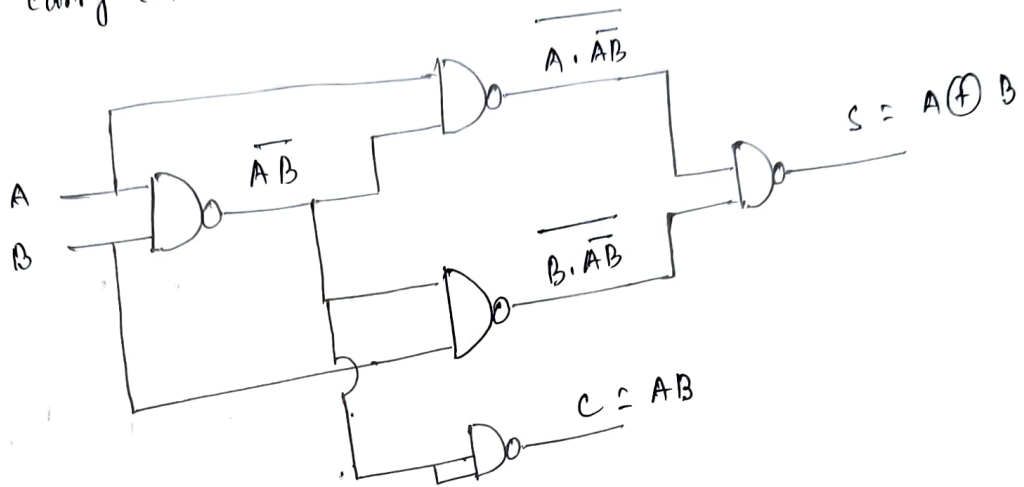


5

For Half adder, we have two o/p i.e. sum & carry

$$\text{sum (s)} = A \oplus B$$

$$\text{carry (c)} = AB$$



6

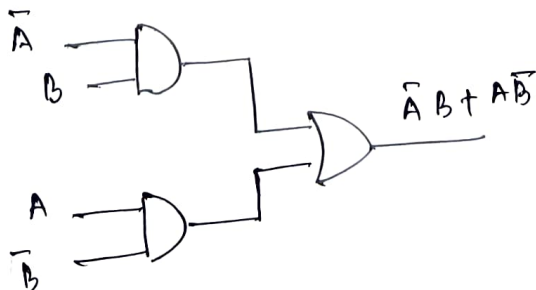
Given $F = \overline{AB + \bar{A}\bar{B}} + \bar{A}B$

so, $F = \overline{AB} \cdot \overline{(\bar{A}\bar{B})} + \bar{A}B$

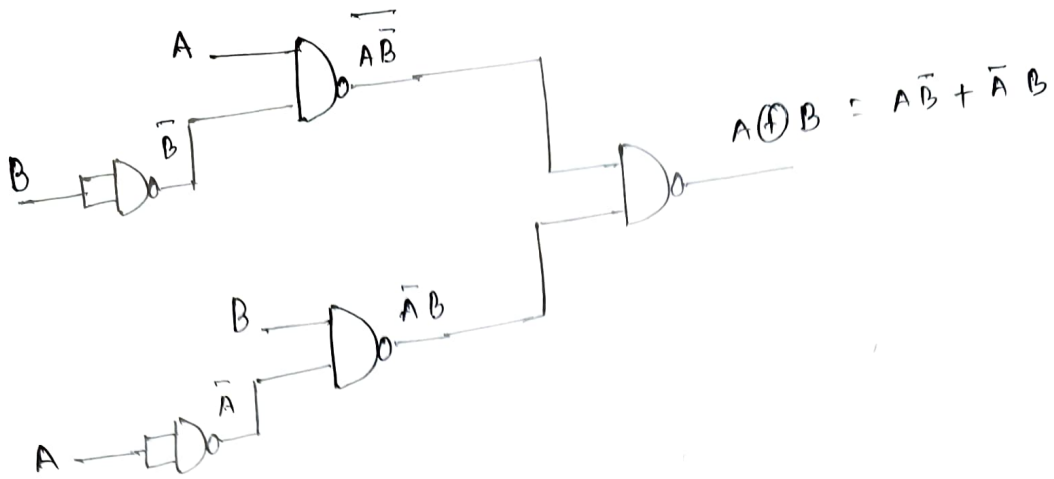
$$F = (\bar{A} + \bar{B})(A + B) + \bar{A}B$$

$$F = \bar{A}B + A\bar{B} = A \oplus B$$

7



⑥



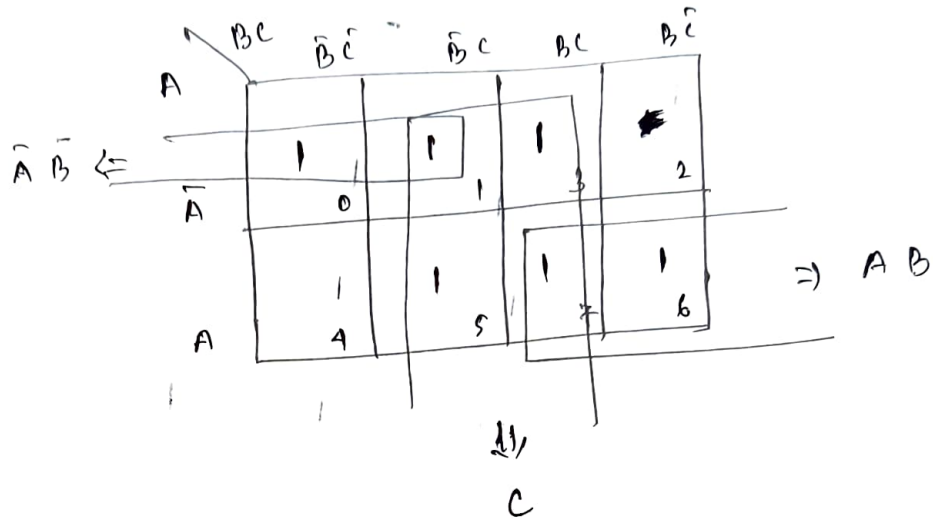
7. a)

A (i/p)	B (i/p)	C (i/p)	F (Bulb' → o/p)
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

⑥ Based on truth table, F can be written as

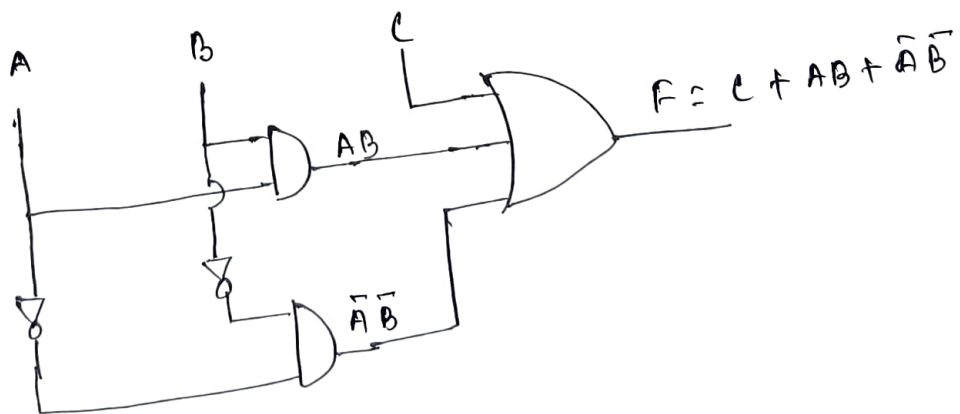
$$F = \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

(0) (1) (3) (5) (6) (7)



So, $F = \bar{A}\bar{B} + AB + C$

⑦ $F = \bar{A}\bar{B} + AB + C$

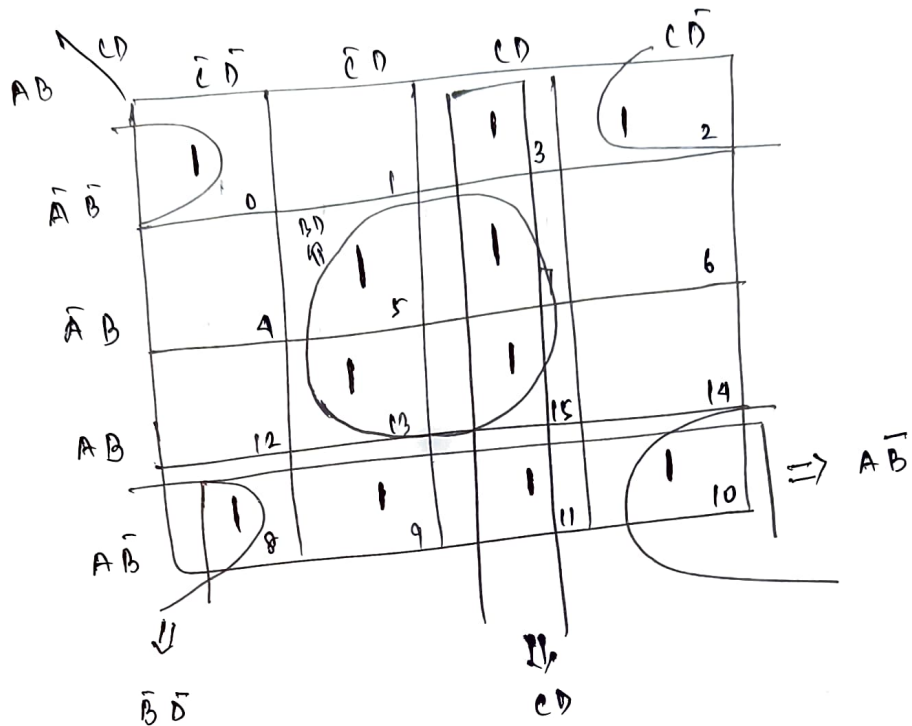


⑧

Given

$$f(A, B, C, D) = \sum (0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$$

Using 4-Variable K-map



$$f(A, B, C, D) = \bar{B}\bar{D} + BD + C\bar{D} + A\bar{B}$$

⑨ Given $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ &

$$d(w, x, y, z) = \Sigma(0, 2, 5)$$

using 4-variable K-map

$w \backslash x \backslash yz$	yz	$\bar{y}z$	$y\bar{z}$	$\bar{y}\bar{z}$
$\bar{w}\bar{x}$	X_0	1	1	X_2
$\bar{w}x$	4	X_5	1	6
$w\bar{x}$	12	13	1	14
wx	8	9	1	10

$\Rightarrow \bar{w}\bar{x}$

w, yz

so,
$$F = \bar{w}\bar{x} + yz$$

⑩

$$F(x, y, z) = \Sigma(2, 3, 4, 5)$$

using 3-variable K-map

$x \backslash yz$	yz	$\bar{y}z$	$y\bar{z}$	$\bar{y}\bar{z}$
$\bar{x}$	0	1	3	2
x	4	5	7	6

$$F = x\bar{y} + \bar{x}y = x \oplus y$$